Tutorial 5

Determinants, eigenvalues and eigenvectors

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1. Let A be a 7×7 matrix, with $\det(A) = 3$. Find
(a) $\det(-4A)$ (b) $\det(A^{-1}A^T)$ (c) $\det(A^4)$ (d) $\det(AA^T)$.

Solution:

(a) $\det(-4A) = (-4)^7 \det(A) = (-4)^7 \cdot 3 = -49152$.

(b) $\det(A^{-1}A^T) = \det(A^{-1}) \cdot \det(A^T) = \frac{1}{\det(A)} \det(A^T) = \frac{1}{\det(A)} \det(A) = 1$.

(c) $\det(A^4) = 3^4 = 81$.

(d) $\det(AA^T) = \det(A) \cdot \det(A^T) = \det(A) \cdot \det(A) = 3^2 = 9$.

2. Let

$$A = \begin{pmatrix} 3 & 0 & -9 & 6 \\ -2 & 5 & 1 & -1 \\ -1 & 7 & 4 & -2 \\ 4 & -1 & 2 & 3 \end{pmatrix}.$$

(a) Perform elementary row operations on matrix A to make all the entries in the first column, except the first entry, to zero.

(b) Use (a) to find $\det(A)$.

Solution:

$$\det(A) = \begin{vmatrix} 3 & 0 & -9 & 6 \\ -2 & 5 & 1 & -1 \\ -1 & 7 & 4 & -2 \\ 4 & -1 & 2 & 3 \end{vmatrix} = 3 \cdot \begin{vmatrix} 1 & 0 & -3 & 2 \\ -2 & 5 & 1 & -1 \\ -1 & 7 & 4 & -2 \\ 4 & -1 & 2 & 3 \end{vmatrix} = 3 \cdot \begin{vmatrix} 1 & 0 & -3 & 2 \\ 0 & 5 & -5 & 3 \\ 0 & 7 & 1 & 0 \\ 0 & -1 & 14 & -5 \end{vmatrix}$$

$$= 3 \cdot \begin{vmatrix} 5 & -5 & 3 \\ 7 & 1 & 0 \\ -1 & 14 & -5 \end{vmatrix} = 3 \cdot 97 = 291.$$

3. Prove or disprove:

(a) For any square matrix A, $\det(-A) = -\det(A)$.

Solution: False: $\det(-I_2) = \det(I_2) = 1$, not -1. In general, $\det(-A) = (-1)^n \det(A)$ if A is an $n \times n$ matrix.

(b) If A is a square matrix, then $\det(A^T A) \geq 0$.

Solution: True. $\det(A^T A) = \det(A^T)\det(A) = (\det(A))^2 \geq 0$, no matter what A is.

(c) If A and B are square matrices of the same size, $\det(A+B) = \det(A) + \det(B)$.

Solution: False. A possible counterexample: Let $A = I_2$, $B = -I_2$. Then $\det(A + B) = 0$, but $\det(A) + \det(B) = 1 + 1 = 2$.

(d) If A is a square matrix such that $A^T = A^{-1}$ then $\det(A) = 1$ or $\det(A) = -1$.

Solution: True. Since $\det(A^T) = \det(A)$ we obtain $\det(A) = \det(A^{-1}) = \frac{1}{\det(A)}$ and so $(\det(A))^2 = 1$, which makes $\det(A) = 1$ or -1. (Note that $\det(A)$ cannot be zero, because A is invertible.)

4. Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 3 & 2 \\ 1 & 2 & 3 \end{pmatrix}$.

(a) Find all the eigenvalues of A.

Solution: To find the eigenvalues of A, you need to solve the equation

$\det(A - \lambda I) = 0$.

$$
\begin{aligned}
0 = \det(A - \lambda I) &= \begin{vmatrix} 1-\lambda & 0 & 0 \\ 1 & 3-\lambda & 2 \\ 1 & 2 & 3-\lambda \end{vmatrix} \\
&= (1-\lambda)\begin{vmatrix} 3-\lambda & 2 \\ 2 & 3-\lambda \end{vmatrix}
\end{aligned}
$$

(expanded along the first row, which is most convenient in this case)

$$
\begin{aligned}
&= (1-\lambda)[(3-\lambda)(3-\lambda) - 4] \\
&= (1-\lambda)[\lambda^2 - 6\lambda + 5] \\
&= (1-\lambda)^2(5-\lambda).
\end{aligned}
$$

So the eigenvalues of A are 1 and 5.

(b) For each eigenvalue you found in (a), find a corresponding eigenvector. [*Hint:* For each eigenvalue, you have to solve a system of linear equations to find a corresponding eigenvector.]

Solution: To find an eigenvector $\boldsymbol{x}$ of A corresponding to the eigenvalue λ, you need to find a *non-zero* solution $\boldsymbol{x}$ of the system of equations

$$(A - \lambda I_3)\boldsymbol{x} = \mathbf{0}.$$

(If your system has only the zero solution, there is something wrong with your calculation of λ.)

- For $\lambda = 5$, $A - \lambda I_3 = \begin{pmatrix} -4 & 0 & 0 \\ 1 & -2 & 2 \\ 1 & 2 & -2 \end{pmatrix}$.

  The solutions of the system

  $$(A - 5I_3)\boldsymbol{x} = \mathbf{0}$$

  are all of the form $s\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$, with $s \in \mathbb{R}$. Any vector of this form with $s \neq 0$ is an eigenvector of A corresponding to the eigenvalue 5.

- For $\lambda = 1$, $A - \lambda I_3 = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 2 & 2 \\ 1 & 2 & 2 \end{pmatrix}$.

  The solutions of the system

  $$(A - 1I_3)\boldsymbol{x} = \mathbf{0}$$

are all of the form $s\begin{pmatrix}-2\\1\\0\end{pmatrix}+t\begin{pmatrix}-2\\0\\1\end{pmatrix}$ with $s,t\in\mathbb{R}$. Any non-zero vector of this form, such as $\begin{pmatrix}-2\\1\\0\end{pmatrix}$ or $\begin{pmatrix}-2\\0\\1\end{pmatrix}$, will be an eigenvector of A corresponding to the eigenvalue 1.

5. Let $A=\begin{pmatrix}7&4&-4\\4&-8&-1\\-4&-1&-8\end{pmatrix}$.

(a) Write down the determinant you have to evaluate to find the eigenvalues of A (there will be λ's in the determinant).

Solution:

$$\det(\lambda I-A)=\begin{vmatrix}\lambda-7&-4&4\\-4&\lambda+8&1\\4&1&\lambda+8\end{vmatrix}.$$

(b) Perform a row operation on this determinant to obtain a zero in the first column.

Solution: Add row 2 to row 3; this row operation does not change the value of the determinant:

$$\begin{aligned}\det(\lambda I-A)&=\begin{vmatrix}\lambda-7&-4&4\\-4&\lambda+8&1\\4&1&\lambda+8\end{vmatrix}\\&=\begin{vmatrix}\lambda-7&-4&4\\-4&\lambda+8&1\\0&\lambda+9&\lambda+9\end{vmatrix}\end{aligned}$$

(c) Then make use of the properties of determinants to find the value of the determinant, in terms of λ.

Solution: Expand the last determinant in (b) above along the last row and use the fact that $\lambda+9$ is a common factor in this row:

$$\begin{aligned}\begin{vmatrix}\lambda-7 & -4 & 4\\ -4 & \lambda+8 & 1\\ 0 & \lambda+9 & \lambda+9\end{vmatrix} &= (\lambda+9)\begin{vmatrix}\lambda-7 & -4 & 4\\ -4 & \lambda+8 & 1\\ 0 & 1 & 1\end{vmatrix}\\ &= (\lambda+9)(\lambda^2-81)\\ &= (\lambda-9)(\lambda+9)^2.\end{aligned}$$

(d) Now find all the eigenvalues of A.

Solution: The eigenvalues of A are the solutions to the equation

$$0 = \det(\lambda I - A) = (\lambda-9)(\lambda+9)^2$$

and therefore are $9, -9, -9$.

(e) For each eigenvalue you have found, find a corresponding eigenvector.

Solution: To find an eigenvector $\boldsymbol{x}$ of A corresponding to the eigenvalue λ, we need to find a *non-zero* solution $\boldsymbol{x}$ of the system of equations

$$(\lambda I_3 - A)\boldsymbol{x} = \mathbf{0}.$$

(If your system has only the zero vector as solution, there is something wrong with your calculation of λ.)

- For $\lambda = 9$,

  $$\lambda I_3 - A = \begin{pmatrix}2 & -4 & 4\\ -4 & 17 & 1\\ 4 & 1 & 17\end{pmatrix}.$$

  We can reduce this matrix to the matrix

  $$\begin{pmatrix}1 & -2 & 2\\ 0 & 1 & 1\\ 0 & 0 & 0\end{pmatrix}.$$

  The solutions of the system

  $$(9I_3 - A)\boldsymbol{x} = \mathbf{0}$$

  are therefore all of the form $s\begin{pmatrix}4\\ 1\\ -1\end{pmatrix}$, with $s \in \mathbb{R}$. Any vector of this form with $s \neq 0$ is an eigenvector of A corresponding to the eigenvalue 9.

- For $\lambda = -9$,

$$\lambda I_3 - A = \begin{pmatrix} -16 & -4 & 4 \\ -4 & -1 & 1 \\ 4 & 1 & -1 \end{pmatrix}.$$

We can reduce this matrix to the matrix

$$\begin{pmatrix} 4 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The solutions of the system

$$(-9I_3 - A)\boldsymbol{x} = \mathbf{0}$$

are therefore all of the form $s\begin{pmatrix} -1 \\ 4 \\ 0 \end{pmatrix} + t\begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix}$, with $s, t \in \mathbb{R}$. Any vector $\mathbf{x} \neq \mathbf{0}$ of this form is an eigenvector of A corresponding to the eigenvalue -9.

6. Let A and P be $n \times n$ matrices, with P invertible.
   (a) Show that A and $P^{-1}AP$ have the same eigenvalues.
   [*Hint:* $I = P^{-1}IP$.]

Solution:

$$\begin{aligned}
\det(P^{-1}AP - \lambda I) &= \det(P^{-1}AP - \lambda P^{-1}IP) \\
&= \det(P^{-1}(A - \lambda I)P) \\
&= \det(P^{-1}) \cdot \det(A - \lambda I) \cdot \det(P) \\
&= \det(P^{-1}) \cdot \det(P) \cdot \det(A - \lambda I) \\
&= \det(P^{-1}P) \cdot \det(A - \lambda I) \\
&= \det(I) \cdot \det(A - \lambda I) \\
&= \det(A - \lambda I)
\end{aligned}$$

Make sure that you can justify each step. Note that in the above we use the fact that real number multiplication is commutative; whereas matrix multiplication of course is not.
Since the eigenvalues of A and $P^{-1}AP$ are respectively the solutions of $\det(A - \lambda I) = 0$ and $\det(P^{-1}AP - \lambda I) = 0$, it follows that the eigenvalues are the same.

(b) If λ is an eigenvalue of A and $\boldsymbol{x}$ a corresponding eigenvector of A, is it also an eigenvector of $P^{-1}AP$? If not, find an eigenvector of $P^{-1}AP$ corresponding to λ.

Solution:

The answer is No.

Claim 1. *Let $\boldsymbol{x}$ is an eigenvector of A corresponding to λ. Then $P^{-1}\boldsymbol{x}$ is an eigenvector of $P^{-1}AP$ corresponding to λ.*

Proof. We need to verify that

$$P^{-1}AP(P^{-1}\boldsymbol{x}) = \lambda(P^{-1}\boldsymbol{x}).$$

Indeed,

$$\begin{aligned} P^{-1}AP(P^{-1}\boldsymbol{x}) &= P^{-1}A(PP^{-1})\boldsymbol{x} \\ &= P^{-1}A\boldsymbol{x} \\ &= P^{-1}\lambda\boldsymbol{x} \text{ (since } \boldsymbol{x} \text{ is eigenvector of } A \text{ corresponding to } \lambda) \\ &= \lambda P^{-1}\boldsymbol{x}. \end{aligned}$$

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